

Lagrange Duality: Basic Notions

Constrained optimization problems:

$$\min_{x \in D} f(x)$$

$$\max_{x \in D} f(x)$$

Lagrange Duality: Basic Notions

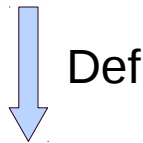
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$$\min_{x \in D} f(x)$$

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If D – convex and :

f – convex



Def

Convex optimization problem

Lagrange Duality: Basic Notions

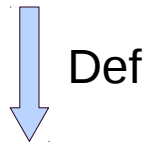
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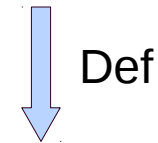
If D – convex and :

f – convex



Convex optimization problem

f – concave



Concave optimization problem

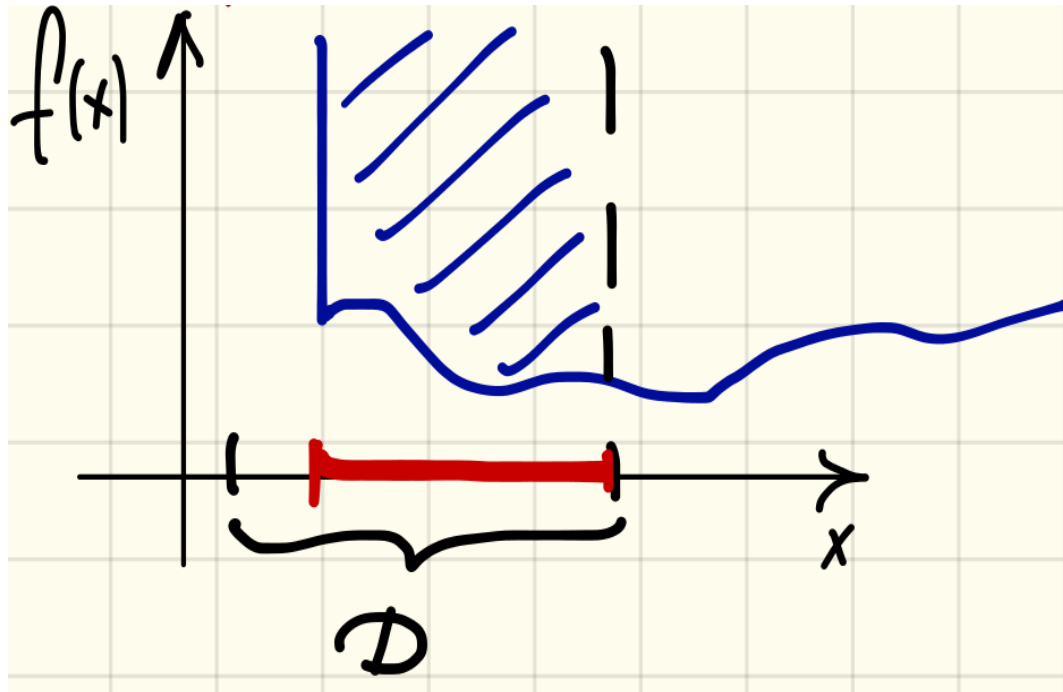
Lagrange Duality: Basic Notions

Constrained optimization problems:

$$\min_{x \in D} f(x)$$

$D \cap \text{dom} f$ – feasible set

$D \cap \text{dom} f \ni x$ – feasible point



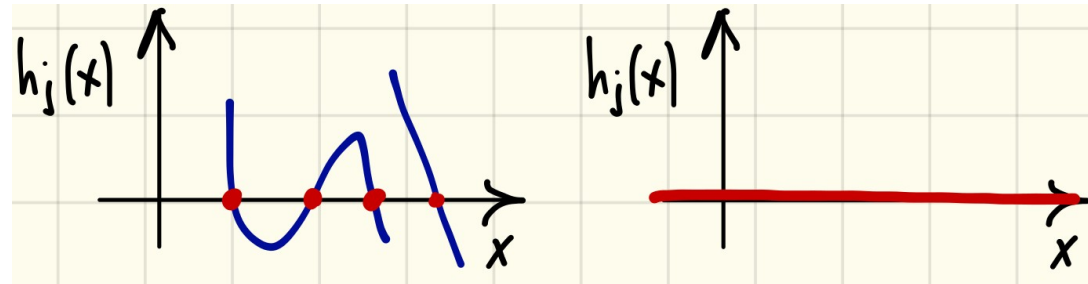
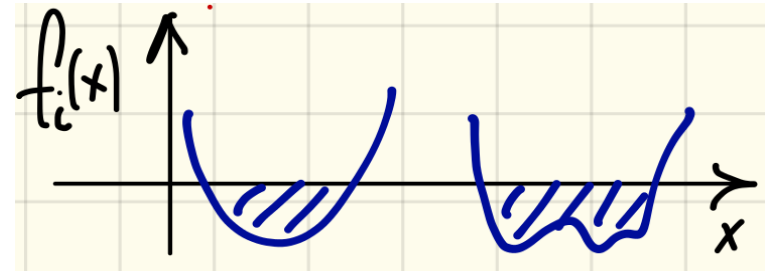
Lagrange Duality: Nonlinear Programming

(Non)linear programming:

$$\min_x f_0(x)$$

$$f_i(x) \leq 0, \quad i = 1, \dots, n$$

$$h_j(x) = 0, \quad j = 1, \dots, m$$



$$f_i: \mathbb{R}^N \rightarrow \mathbb{R} \cup \{\infty, -\infty\}$$

$$h_j: \mathbb{R}^N \rightarrow \mathbb{R} \cup \{\infty, -\infty\}$$

Lagrange Duality: Lagrangian

Lagrangian:

$$L(x, \lambda, \nu) := f_0(x) + \sum_{i=1}^n \lambda_i f_i(x) + \sum_{j=1}^m \nu_j h_j(x)$$

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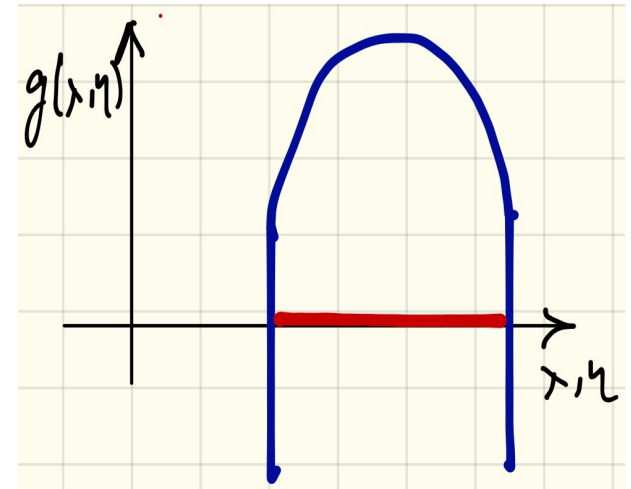
Lagrange Duality: Lagrangian Dual

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Lagrangian dual (objective):

$$g(\lambda, \nu) := \inf_{x \in D} L(x, \lambda, \nu)$$



Lagrange Duality: Lagrangian Dual

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Lagrangian dual (objective):

$$g(\lambda, \nu) := \inf_x L(x, \lambda, \nu)$$

Lagrangian dual (problem):

$$\max_{\lambda, \nu} g(\lambda, \nu)$$

$$g(\lambda, \nu) > -\infty$$

$$\lambda \geq 0$$

Lagrangian Dual of LP

$$\min_x \langle c, x \rangle$$

$$Ax = b$$

$$x \geq 0.$$

Lagrangian Dual of LP

$$\begin{array}{ccc} \min_x \langle c, x \rangle & & \min_x \langle c, x \rangle \\ Ax = b & \longrightarrow & Ax - b = 0 \\ x \geq 0. & & -x \leq 0. \end{array} \quad \begin{array}{c} \nu \\ \lambda \end{array}$$

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Lagrangian:

$$L(x, \lambda, \nu) = \langle c, x \rangle + \langle \nu, (Ax - b) \rangle - \langle \lambda, x \rangle$$

Lagrangian Dual of LP

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Dual:

$$g(\lambda, \nu) = \inf_x L(x, \lambda, \nu) = \inf_x \left(-\langle \nu, b \rangle + \langle c - \lambda + A^\top \nu, x \rangle \right)$$

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Lagrangian Dual of LP

Dual:

$$g(\lambda, \nu) = -\langle \nu, b \rangle + \inf_x (\langle c - \lambda + A^\top \nu, x \rangle)$$

$$\inf_x (\dots) > -\infty \quad \Rightarrow \quad c - \lambda + A^\top \nu = 0$$

Lagrangian Dual of LP

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Dual problem:

$$\max_{\lambda, \nu} g(\lambda, \nu)$$

$$\max_{\nu, \lambda} -\langle \nu, b \rangle$$

$$g(\lambda, \nu) > -\infty$$



$$c - \lambda + A^\top \nu = 0,$$

$$\lambda \geq 0$$

$$\lambda \geq 0$$

Lagrangian Dual of LP

Dual:

$$g(\lambda, \nu) = -\langle \nu, b \rangle + \inf_x (\langle c - \lambda + A^\top \nu, x \rangle)$$

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Dual problem:

$$\max_{\nu, \lambda} -\langle \nu, b \rangle$$

$$c - \lambda + A^\top \nu = 0, \quad \lambda \geq 0 \quad \Rightarrow \quad \begin{aligned} &\max_{\nu} -\langle \nu, b \rangle \\ &c + A^\top \nu \geq 0 \end{aligned}$$

Lagrangian Dual of LP

Dual:

$$g(\lambda, \nu) = -\langle \nu, b \rangle + \inf_x (\langle c - \lambda + A^\top \nu, x \rangle)$$

$$\inf_x (\dots) > -\infty \quad \Rightarrow \quad c - \lambda + A^\top \nu = 0$$

Dual problem:

$$\max_{\nu, \lambda} -\langle \nu, b \rangle$$

$$\begin{array}{ccccc} c - \lambda + A^\top \nu = 0, & \xrightarrow{\quad} & \max_{\nu} -\langle \nu, b \rangle & \xrightarrow[\nu \rightarrow -\nu]{} & \max_{\nu} \langle \nu, b \rangle \\ \lambda \geq 0 & & c + A^\top \nu \geq 0 & & A^\top \nu \leq c \end{array}$$

Lagrangian Dual of LP

Any LP can be represented in

standard

and

canonical forms

$$\min_x \langle c, x \rangle$$

$$\max_{\nu} \langle \nu, b \rangle$$

$$Ax = b$$

$$A^{\top} \nu \leq c$$

$$x \geq 0.$$

primal

dual

Dual problem to LP is LP itself

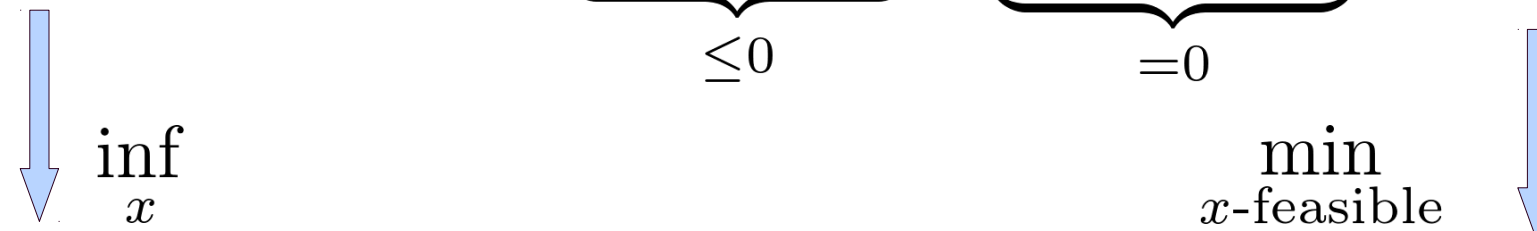
Dual Bound, Weak Duality

Theorem (Weak duality): For any $\lambda \geq 0$ and any ν it holds

$$g(\lambda, \nu) \leq f_0(x^*) = p^*$$

For any feasible x :

$$L(x, \lambda, \nu) = f_0(x) + \underbrace{\sum_{i=1}^n \lambda_i f_i(x)}_{\leq 0} + \underbrace{\sum_{j=1}^m \nu_j h_j(x)}_{=0} \leq f_0(x)$$



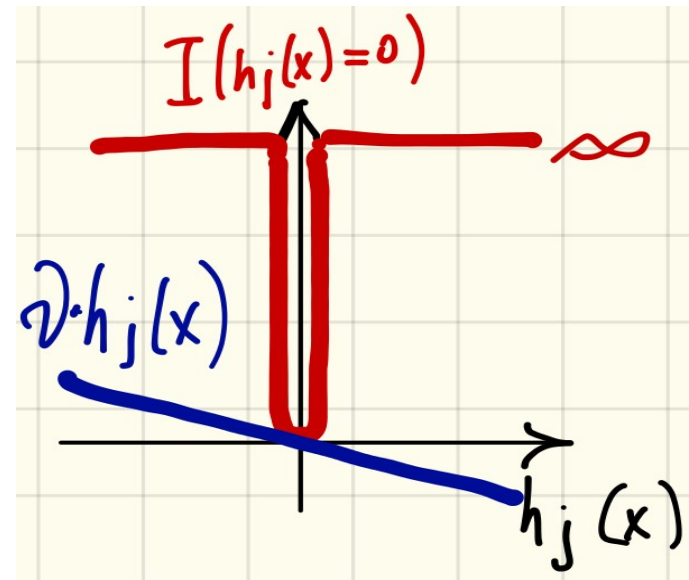
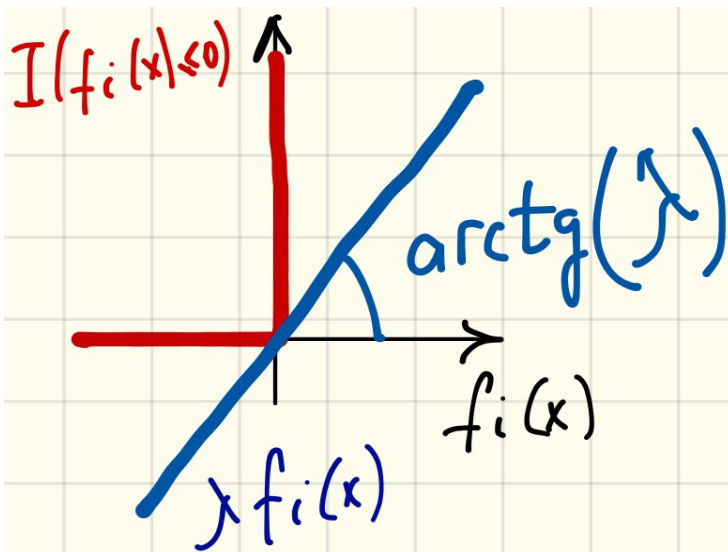
$\inf_x$ $\min_{x\text{-feasible}}$ $\downarrow$

$g(\lambda, \nu)$ $f_0(x^*)$

Lagrangian as Lower Dual Bound

$$\tilde{f}(x) = f_0(x) + \sum_{i=1}^n I(f_i(x) \leq 0) + \sum_{j=1}^m I(h_j(x) = 0)$$

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^n \lambda_i f_i(x) + \sum_{j=1}^m \nu_j h_j(x)$$



Strong Duality

Theorem (Strong duality): Let f_i be convex and h_j linear. Then the optimal dual value is equal to optimal primal one.

$$\begin{array}{ll} \min_x f_0(x) & = \max_{\lambda, \nu} g(\lambda, \nu) \\ f_i(x) \leq 0, \quad i = 1, \dots, n & g(\lambda, \nu) > -\infty \\ h_j(x) = 0, \quad j = 1, \dots, m & \lambda \geq 0 \end{array}$$

Basically if the primal problem is convex – the strong duality holds.

